Design of a Broadband Polarization Controller Based on Silicon Nitride-Loaded Thin-Film Lithium Niobate

JIŘÍ ČTYROKÝ,[1,2,*] JIŘÍ PETRÁČEK,[3,4] IVAN RICHTER,[2] AND VLADIMÍR KUZMIAK[1]

[1]*Institute of Photonics and Electronics, Czech Academy of Sciences, Chaberská 57, 18251 Prague, Czech Republic*
[2]*Faculty of Nuclear Sciences and Physical Engineering, Czech Technical University in Prague, Břehová 7, 11519 Prague 1, Czech Republic*
[3]*Faculty of Mechanical Engineering, Brno University of Technology, Technická 2896/2, 61669 Brno, Czech Republic*
[4]*Central European Institute of Technology, Brno University of Technology, Purkyňova 656/123, 61200 Brno, Czech Republic*
**ctyroky@ufe.cz*

Abstract: A novel design of a polarization controller based on "etch-less" Si_3N_4-loaded thin film $LiNbO_3$ is described. Broadband operation in the spectral range between 1.45 and 1.65 µm is achieved by using a mode evolution TM/TE splitter/converter, two mode evolution 3-dB couplers, and two electro-optic phase shifters. Numerical simulations show that a single TE-mode output can be adjusted by applying control voltages lower than 10 V for an arbitrary input polarization state.

1. Introduction

Polarization controllers are important components of optical systems. While the polarization manipulation in an isotropic medium such as a free space can be performed by the combination of suitably adjusted $\lambda/4$ and $\lambda/2$ phase plates, in generally birefringent integrated-optic waveguides that support the (quasi)-TE and (quasi)-TM modes, the situation is somewhat different. For complete polarization control, the ability of the TE/TM conversion and controllable phase shifts are both required. The early guided-wave devices for polarization management [1-8] were mostly based on metal-diffused $LiNbO_3$ waveguides because of their strong electro-optic properties enabling easy phase shifting. The realization of the TE/TM conversion in such waveguides was based on the coupling of the TE and TM modes; this may be challenging since the modes are nearly linearly and mutually perpendicularly polarized, and they also differ in the propagation constants due to the combined material and waveguide birefringence. The TE/TM coupling was typically mediated by an off-diagonal permittivity component introduced electro-optically, and the phase matching of the TE and TM modes was attained either by a periodic structure of electrodes, or by choosing the direction of propagation close to the optic axis of the $LiNbO_3$ crystal.

Recent development of the thin-film lithium niobate technology (called also lithium niobate on insulator–LNOI) [9] has brought a disruptive change in the design of integrated-optic guided-wave $LiNbO_3$ devices. This novel approach has led to the design of more compact devices with smaller foot size and lower operation voltages. In order to localize light in the lateral direction, etched rib or dielectric-loaded channel waveguides are used. Dispersion properties of such waveguides are more complicated than properties of traditional titanium-diffused $LiNbO_3$ waveguides since the natural birefringence of the $LiNbO_3$ crystal is now combined with the waveguide birefringence of the $LiNbO_3$ crystal slab. As a result, one has to mitigate possible lateral leakage of TM polarized modes due to coupling with the TE polarized $LiNbO_3$ slab mode [10, 11], in analogy with the silicon on insulator (SOI) waveguides [12-15].

Nevertheless, various designs of polarization rotators and splitters on thin-film $LiNbO_3$ platform have been recently published [16-21].

A complete electro-optic polarization control in a thin-film $LiNbO_3$ device has been recently described in a very comprehensive paper [22]. Our present work is devoted to a similar though a more difficult problem–to design a broadband device operating in the wavelength range from 1450 to 1650 nm, capable of transforming an arbitrary polarized input wave into a TE polarized output mode. In order to avoid uneasy etching of a $LiNbO_3$ slab, we consider waveguides laterally confined by silicon nitride loading stripes, similarly as in [23]. For efficient electro-optic control, the X-cut $LiNbO_3$ film was chosen. Analogously to [22], the device is composed of a polarization mode splitter/converter, two 3-dB couplers, and two electro-optic phase shifters. To ensure broadband operation, adiabatic mode evolution components are applied. Their design, inspired to some extent by [21, 24-26], will be described in detail below. As a necessary pre-requisite of these designs, the properties of the silicon nitride-loaded thin film $LiNbO_3$ waveguides will be briefly reviewed. For numerical simulations, COMSOL Multiphysics mode solver [27] and our proprietary 3-D Fourier modal method (FMM) [28] were used.

2. Properties of Si_3N_4-loaded X-cut thin-film $LiNbO_3$ waveguides

The cross-section of the silicon nitride-loaded thin-film $LiNbO_3$ waveguide considered is shown in Fig. 1(a). The X-cut $LiNbO_3$ slab thickness is 400 nm, the silicon nitride stripe thickness is 300 nm. Light propagates along the crystal axis Y.

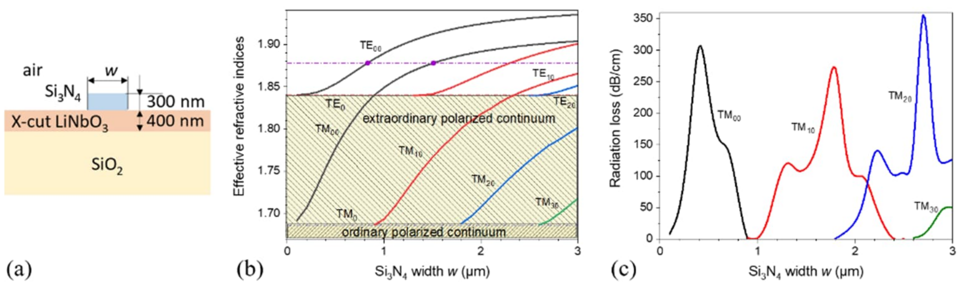

Fig. 1. (a) Cross-section of the Si_3N_4-loaded thin-film $LiNbO_3$ waveguide. (b) Dependence of the effective refractive indices of supported modes on the silicon nitride stripe width *w* at the wavelength of 1550 nm. (c) Radiation loss of the TM modes due to coupling with the TE-polarized mode of the $LiNbO_3$ slab planar waveguide. The data shown in (b) and (c) were calculated using COMSOL Multiphysics [27].

Fig. 1(b) shows the dependences of the effective refractive indices of the TE_{m0} and TM_{m0} modes of the channel waveguide on the silicon nitride stripe width *w* at the wavelength of 1550 nm. Despite the fact that the extraordinary refractive index of the $LiNbO_3$ crystal is lower than the ordinary one and the TE_{m0} modes are predominantly extraordinary polarized, their effective refractive indices are higher than those of the TM_{m0} modes of the same (lateral) order *m* due to the waveguide birefringence of the planar $LiNbO_3$ thin film waveguide supporting the TE_0 and TM_0 modes. The TM_{m0} modes with the effective refractive indices lower than the effective refractive index of the TE_0 mode, which takes place for narrower widths *w*, are extremely leaky due to the coupling with the TE_0 mode via their minority field components parallel to the interfaces. Their radiation losses are shown in Fig. 1(c). From Fig. 1(b) it is evident that we can choose a particular pair of stripe widths for which the corresponding effective refractive indices of the TE_{00} and TM_{00} modes are identical (an example of such pair is indicated by dots on dispersion curves of the modes). This feature will be later utilized in the design of the TM/TE polarization converter as the key component of the polarization controller. The electric field distributions of the relevant modes for the stripe widths 810 nm and 1500 nm are shown in Fig. 2. Note that the mode fields are well localized in the transverse direction, and that the TE mode fields are concentrated mainly in the $LiNbO_3$ slab which is important for efficient electro-optic control. The design does not make use of the bound modes in the continuum [29] which makes it independent of the limitations to the “magic widths” of waveguides and of reduced electro-optic efficiency of phase shifters [30].

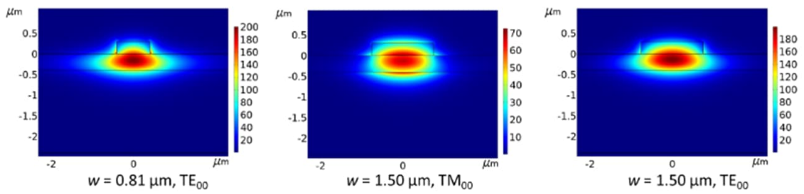

Fig. 2. Distributions of the dominant electric field components (in relative units) in the waveguides with silicon nitride stripe widths of 0.81 µm and 1.5 µm.

3. Polarization controller

The configuration of the designed broadband polarization controller is schematically shown in Fig. 3. It consists of the input taper which prevents the bottom (narrower) waveguide from excitation and transforms the waveguide widths to the required input widths of the wideband adiabatic TM/TE splitter/converter. The upper waveguide is generally excited with the superposition of TE_{00} and TM_{00} modes with arbitrary amplitude ratio and relative phase shift. The adiabatic TM/TE splitter/converter transforms the input TM_{00} mode of the upper (wider) waveguide into the TE_{00} mode of the lower (narrower) waveguide. The aim of the first phase shifter is to adjust the relative phase shift between the input ports of the first broadband adiabatic 3-dB coupler to equalize the mode amplitudes at its output ports. The second phase shifter is used to adjust the relative phase of the input ports of the second 3-dB coupler in order to minimize the output from the bottom port and maximize the amplitude of the TE_{00} mode leaving the upper output port. To reduce the device dispersion, the 3-dB couplers are mutually horizontally flipped.

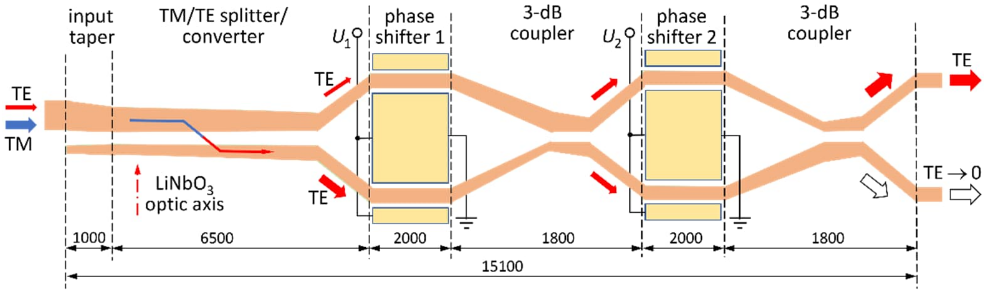

Fig. 3. Schematic configuration of the polarization controller. The length dimensions are in micrometers.

In the next sessions, the configurations of the broadband adiabatic components and phase shifters are described in detail.

4. Broadband TM/TE splitter/converter

Design of adiabatic mode evolution components is not straightforward since adiabaticity is, in fact, an abstraction that can be achieved only approximately. We adopted the procedure similar but not identical to that used in [21]. The key step in the design is the calculation of parameters corresponding to the hybridization of the supermodes of the pair of the waveguides [21], i.e., the situation when the difference between the effective refractive indices of the supermodes is the smallest and the polarization of the supermodes is strongly hybrid. We fixed the edge-to-edge gap between the waveguides to 0.5 µm, which ensures sufficient mode coupling without excessive fabrication demands. Then, based on numerical simulations, we set the widths w_1 and w_2 of the pair of the broader and narrower waveguides at the input to $w_{1,\mathrm{in}} = 1.45\ \mu\mathrm{m}$ and $w_{2,\mathrm{in}} = 0.7\,\mu\mathrm{m}$, and at the output to $w_{1,\mathrm{out}} = 1.32\ \mu\mathrm{m}$ and $w_{2,\mathrm{out}} = 0.9\ \mu\mathrm{m}$, respectively. The input and output values correspond to the mode hybridization for the wavelength of $\lambda_{\max} = 1.65\ \mu\mathrm{m}$ and $\lambda_{\min} = 1.45\ \mu\mathrm{m}$, respectively. Simulations also show that the optimum waveguide widths depend approximately linearly on the wavelength:

$$w_j(\lambda) \approx \left[\left(\lambda - \lambda_{\min}\right) w_{j,\mathrm{in}} + \left(\lambda_{\max} - \lambda\right) w_{j,\mathrm{out}}\right] \big/ \left(\lambda_{\max} - \lambda_{\min}\right),\ j = 1,2,\ \lambda_{\min} \le \lambda \le \lambda_{\max}. \quad (1)$$

The optimum length of the *uniform* coupler with the optimum waveguide widths $w_1(\lambda)$ and $w_2(\lambda)$ leading to hybridization is $L(\lambda) = \lambda \big/ \left[2\left(N_0(\lambda) - N_1(\lambda)\right)\right]$, where N_0 and N_1 are the fundamental and the higher-order effective refractive indices of the hybridized modes of the waveguide pair, respectively. Numerical simulations confirmed the expectation that the function $L(\lambda)$ is approximately exponentially dependent on the wavelength, $L(\lambda) \sim \exp(-\alpha\lambda)$. Specifically, for the chosen parameters, $L(\lambda_{\min}) \approx 2.85\, L(\lambda_{\max})$. To enable numerical simulation of the adiabatic polarization splitter/converter with our FMM numerical code [28] (written in cartesian coordinates), we discretized the wavelength range into p wavelengths,

$$\lambda_l = \lambda_{\max} - (l-1)(\lambda_{\max} - \lambda_{\min})/(p-1),\ l = 1,2,\ldots p, \quad (2)$$

and approximated the smoothly tapered adiabatic section with p longitudinally uniform sections with waveguide widths $w_j(\lambda_l)$, each of which was optimized for mode hybridization at λ_l. The lengths L_l of these sections were chosen to exponentially increase according to the rule

$$L_l = qL_{l-1},\ \ L_1 = L_c\,(q-1)\big/(q^p - 1),\ q = \left[L(\lambda_{\min})/L(\lambda_{\max})\right]^{1/(p-1)},\ l = 2,3,\ldots p, \quad (3)$$

where L_c is the total length of the adiabatic section of the polarization splitter/converter. Repeated simulations showed that increasing the number of sections p above 50 did not lead to noticeable change of the results. The total length was then fixed at $L_c = 6000\ \mu\mathrm{m}$, which led to the smallest device footprint without compromising the TE/TM conversion efficiency.

The waveguide bends at the output of the adiabatic section ensure that the waveguides are separated, to prevent from further unwanted mode coupling, and their widths are simultaneously tapered to the identical width $w = 0.85\,\mu\mathrm{m}$ used in the electro-optic phase shifters. The shapes of these bends were designed as 5^{th}-order polynomials which allow for smooth waveguide tapering without discontinuities in the waveguide directions and curvatures. The length of the bends of 500 µm and the final edge-to-edge separation of the waveguides of 8 µm were chosen as a trade-off between the bend losses and the footprint size. The configuration of the adiabatic TM/TE mode splitter/converter and the electric field distributions under TM and TE mode excitations at different wavelengths are shown in Fig. 4. Calculated propagation and conversion losses were lower than 0.2 dB in the spectral range from 1.45 µm to 1.65 µm.

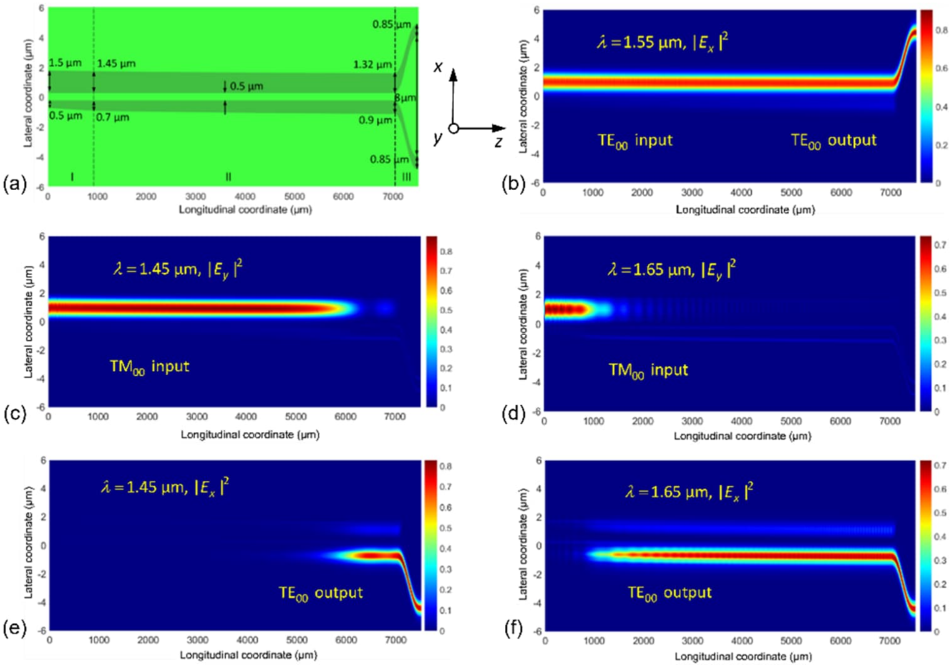

Fig. 4. (a) Configuration of the adiabatic mode evolution TM/TE splitter/converter: I–input taper, II–adiabatic section, III–output bends; (b) - (f) –electric field distributions inside the TM/TE splitter/converter (in relative units): (b) $\lambda = 1.55$ µm, TE$_{00}$ input, $|E_x|^2$; (c) $\lambda = 1.45$ µm, TM$_{00}$ input, $|E_y|^2$; (d) same for $\lambda = 1.65$ µm; e) $\lambda = 1.45$ µm, TM$_{00}$ input, $|E_x|^2$; (f) same for $\lambda = 1.65$ µm.

5. Broadband 3-dB couplers

Probably the most straightforward design of a broadband 3-dB coupler is based on the concatenation of single-mode asymmetric and symmetric Y-junctions connected with the two-mode section, as shown in Fig. 5(a). Since the correctly designed asymmetric Y-junction behaves adiabatically as a mode splitter/divider, light coupled into the wider input waveguide of the asymmetric Y-junction excites in the common two-mode section the symmetric mode while coupling into the narrower waveguide excites in the common section the antisymmetric mode. In both cases, light power is then divided symmetrically in the symmetric Y-junction into the output ports, resulting in 3-dB coupling. Reciprocity of the device guarantees the 3-dB coupling in the opposite direction, too.

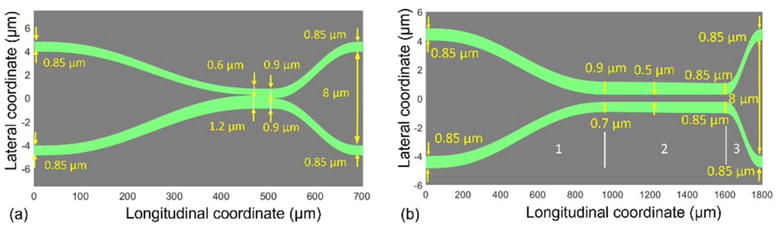

Fig. 5. (a) 3-dB coupler composed of the concatenation of asymmetric and symmetric Y-junctions; (b) modified design: 1–input bends, 2–adiabatic mode evolution section, 3–symmetric output bends.

Numerical simulation of this device unfortunately revealed that at the shortest wavelength of 1.45 µm, the central section supported three modes, which lead to wavelength-dependent radiation loss. Moreover, sharp tips at the waveguide branching make the fabrication of this device challenging. Therefore, we modified the design by separating the waveguides in the adiabatic mode evolution section, as shown in Fig. 5(b). All bends are also shaped as 5th-order polynomials. Broadband operation of the modified coupler is demonstrated by the field distributions shown in Fig. 6. Numerical simulation using the FMM [28] shows the propagation and splitting loss lower than 0.05 dB and the splitting asymmetry lower than 0.2 dB in the spectral range considered.

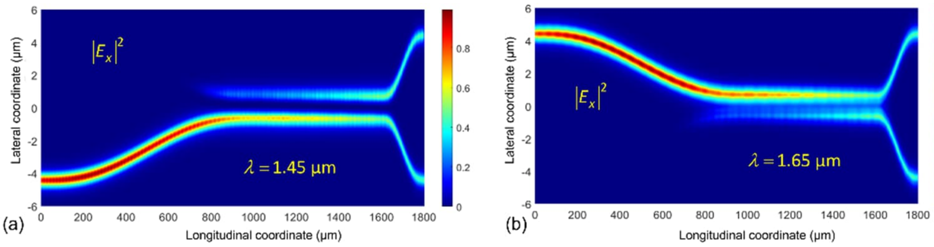

Fig. 6. $|E_x|^2$ field distributions in the 3-dB coupler (in relative units) at the wavelengths of (a) 1.45 μm and (b) 1.65 μm.

6. Electro-optic phase shifters

For the electro-optic phase shifters, a standard push-pull configuration is implemented. The X-cut Y-propagation arrangement allows utilization of the strong electro-optic coefficient r_{33} of the $LiNbO_3$ slab. Electrodes are designed as 6 μm wide, separated by 1 μm from the Si_3N_4 stripe, composed of the 150 nm SiO_2 buffer covered with 200 nm of gold. The buffer is used to suppress light absorption in metal electrodes. Numerical simulations by COMSOL Multiphysics 6.1 [27] showed that the voltage of 1 V applied to the electrodes excites nearly uniform horizontal electric field intensity of 2×10^5 V/m in the $LiNbO_3$ slab. Numerical simulations predicted optical absorption loss due to electrodes lower than 0.25 dB/cm. The absorption loss in both phase shifters, the total electrode length of which is 4 mm, is thus of the order of 0.1 dB. The half-wave voltage U_π of the phase shifters was estimated to be lower than 10 V for any wavelength within the wavelength band considered. The distribution of the DC electric field calculated with COMSOL is shown in Fig. 7.

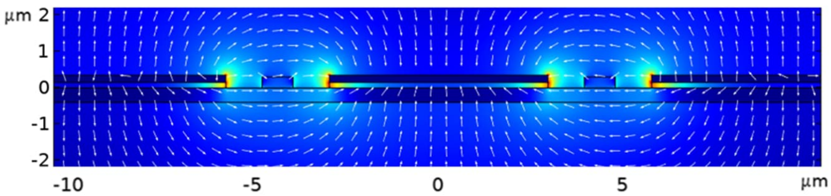

Fig. 7. Distribution of the DC electric field in the phase shifters. Arrows indicate the direction of the electric field.

7. Operation of the polarization controller

The components of the controller–the TM/TE splitter/converter, the phase shifters and the 3dB couplers–can be considered as linear four-ports, as it is shown in Fig. 8.

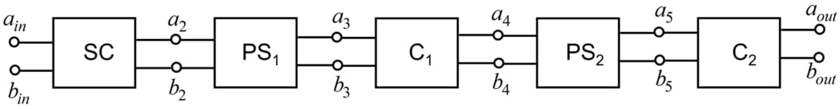

Fig. 8. Representation of the controller by a cascade of four-ports. SC—TM/TE splitter/converter, PS_1, PS_2—phase shifters, C_1, C_2—3-dB couplers, a , b –complex amplitudes in the pertinent ports.

The complex amplitudes a_{in} and b_{in} correspond to the input polarizations TE and TM, respectively, while the amplitudes a_m and b_m, $m = 2,\ldots 5$ a_{out}, b_{out} represent complex amplitudes of the TE modes in the pertinent ports.

Numerical simulations showed that back-reflections from the individual components are negligible. Each component can thus be represented by the corresponding 2×2 transfer matrix. Neglecting losses, the matrices gain very simple forms shown in (4):

$$\mathbf{T}_{SC} = \begin{pmatrix} e^{i\varphi^{TE}} & 0 \\ 0 & e^{i\varphi^{TM}} \end{pmatrix}, \mathbf{T}_{PS_1} = \begin{pmatrix} e^{i\varphi_1} & 0 \\ 0 & e^{-i\varphi_1} \end{pmatrix}, \mathbf{T}_{C_1} = \frac{1}{\sqrt{2}}\begin{pmatrix} e^{i\varphi_w} & e^{i\varphi_n} \\ e^{i\varphi_w} & -e^{i\varphi_n} \end{pmatrix},$$
$$\mathbf{T}_{PS_2} = \begin{pmatrix} e^{i\varphi_2} & 0 \\ 0 & e^{-i\varphi_2} \end{pmatrix}, \mathbf{T}_{C_2} = \frac{1}{\sqrt{2}}\begin{pmatrix} -e^{i\varphi_n} & e^{i\varphi_w} \\ e^{i\varphi_n} & e^{i\varphi_w} \end{pmatrix}. \tag{4}$$

Here φ^{TE} and φ^{TM} are phase shifts due to propagation of the TE and TM polarized modes within the splitter/converter, respectively, $\varphi_1 = (\pi/2)(U_1/U_\pi)$, $\varphi_2 = (\pi/2)(U_2/U_\pi)$, U_1 and U_2 are control voltages of the phase shifters PS_1 and PS_2, respectively, U_π is their half-wave voltage, and φ_w φ_n are phase shifts of the modes entering the wider and narrower input ports of the 3-dB couplers, respectively. A straightforward analysis shows that for arbitrary input complex amplitudes a_{in} and b_{in} the voltage U_1 can be always adjusted within the limits $|U_1| \le U_\pi/2$ in such a way that the absolute values of the complex amplitudes a_4 and b_4 are equal. The optimum voltage U_1 depends only on the relative phase shift between the amplitudes of the input TE and TM modes a_{in} and b_{in}, regardless of their amplitude ratio. The voltage U_2 can be adjusted within the limits $|U_2| \le U_\pi$ to minimize the output amplitude b_{out} and to maximize the amplitude a_{out}. Both input TE_{00} and TM_{00} modes are thus coherently combined into the single output TE_{00} mode. Obviously, the interpretation of the operation of the polarization controller using the Poincaré sphere as in [22] can be used as well.

8. Conclusions

A numerical proof of principle of an efficient broadband electro-optic polarization controller based on silicon nitride-loaded X-cut thin-film lithium niobate platform is presented. The device is designed to provide a single-mode TE polarization output from an arbitrary input polarization in the wavelength range of 1.45 to 1.65 μm without the need of uneasy etching of the $LiNbO_3$ crystal. Original designs of the key adiabatic broadband components—the polarization splitter/converter and the 3-dB coupler—are described. Conventional two-electrode phase shifters utilizing the high electro-optic coefficient r_{33} of the $LiNbO_3$ crystal are designed for phase tuning. Numerical simulations indicate that the required operation voltage of both phase shifters required for endless polarization control should be lower than 10 V.

The cumulative insertion loss of the controller obtained as a sum of losses of individual components was found to be about 0.4 dB. Taking into account the typical background propagation loss in the silicon nitride-loaded LNOI of about 1 dB/cm, the total on-chip loss of the 1.5-cm long device can be estimated as lower than 2 dB.

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Funding. Akademie věd České republiky (RVO 67985882); České vysoké učení technické (RVO 608407700); Ministerstvo školství, mládeže a tělovýchovy (CZ.02.1.01/0.0/0.0/16_019/0000778); Vysoké učení technické v Brně (FSI-S-23-8389).

References

1. R. C. Alferness and L. L. Buhl, "Waveguide electrooptic polarization transformer," Appl. Phys. Lett. **40**(10), 861-862 (1982).
2. S. Thaniyavarn, "Wavelength-independent, optical-damage-immune $LiNbO_3$ TE-TM mode converter," Opt. Lett. **11**(1), 39-41 (1986).
3. F. Heismann and R. C. Alferness, "Wavelength-tunable electrooptic polarization conversion in birefringent wave-guides," IEEE J. Quantum Electron. **24**(1), 83-93 (1988).
4. R. Noé, H. Heidrich, and D. Hoffmann, "Automatic endless polarization control with integrated-optical. Ti:LiNbO3 polarization transformers," Opt. Lett. **13**(6), 527-529 (1988).
5. F. Heismann and M. S. Whalen, "Broad-band reset-free automatic polarization controller," Electron. Lett. **27**(4), 377-379 (1991).
6. F. Heismann and M. S. Whalen, "Fast automatic polarization control-system," IEEE Phot. Tech. Lett. **4**(5), 503-505 (1992).
7. Y. Ping, O. Eknoyan, and H. F. Taylor, "Electro-optic polarization converter with programmable spectral output in lithium niobate," Opt. Commun. **278**(2), 307-311 (2007).
8. B. Koch, R. Noe, D. Sandel, and V. Mirvoda, "Versatile endless optical polarization controller/tracker/demultiplexer," Opt Express **22**(7), 8259-8276 (2014).
9. A. Boes, B. Corcoran, L. Chang, J. Bowers, and A. Mitchell, "Status and Potential of Lithium Niobate on Insulator (LNOI) for Photonic Integrated Circuits," Laser & Photonics Reviews **12**(4), 1700256 (2018).
10. S. T. Peng and A. A. Oliner, "Guidance and leakage properties of a class of open dielectric wave-guides: Part I–Mathematical formulations," IEEE Trans. Microwave Theory Tech. **29**(9), 843-855 (1981).
11. A. A. Oliner, S. T. Peng, T. I. Hsu, and A. Sanchez, "Guidance and leakage properties of a class of open dielectric wave-guides: Part II–New physical effects," IEEE Trans. Microwave Theory Tech. **29**(9), 855-869 (1981).
12. M. A. Webster, R. M. Pafchek, A. Mitchell, and T. L. Koch, "Width Dependence of Inherent TM-Mode Lateral Leakage Loss in Silicon-On-Insulator Ridge Waveguides," IEEE Phot. Tech. Lett. **19**(6), 429-431 (2007).
13. M. Koshiba, K. Kakihara, and K. Saitoh, "Reduced lateral leakage losses of TM-like modes in silicon-on-insulator ridge waveguides," Opt. Lett. **33**(17), 2008-2010 (2008).
14. T. G. Nguyen, R. S. Tummidi, T. L. Koch, and A. Mitchell, "Rigorous Modeling of Lateral Leakage Loss in SOI Thin-Ridge Waveguides and Couplers," IEEE Phot. Tech. Lett. **21**(7), 486-488 (2009).
15. T. G. Nguyen, A. Boes, and A. Mitchell, "Lateral Leakage in Silicon Photonics: Theory, Applications, and Future Directions," IEEE Journal of Selected Topics in Quantum Electronics **26**(2), 1-13 (2020).
16. H. N. Xu, D. X. Dai, L. Liu, and Y. C. Shi, "Proposal for an ultra-broadband polarization beam splitter using an anisotropy-engineered Mach-Zehnder interferometer on the x-cut lithium-niobate-on-insulator," Opt. Express **28**(8), 10899-10908 (2020).
17. Z. X. Chen, J. W. Yang, W. H. Wong, E. Y. B. Pun, and C. Wang, "Broadband adiabatic polarization rotator-splitter based on a lithium niobate on insulator platform," Photonics Research **9**(12), 2319-2324 (2021).
18. X. Han, Y. Jiang, A. Frigg, H. Xiao, P. Zhang, T. G. Nguyen, A. Boes, J. Yang, G. Ren, Y. Su, A. Mitchell, and Y. Tian, "Mode and Polarization-Division Multiplexing Based on Silicon Nitride Loaded Lithium Niobate on Insulator Platform," Laser & Photonics Reviews **16**(1), (2021).
19. X. Wang, A. Pan, T. Li, C. Zeng, and J. Xia, "Efficient polarization splitter-rotator on thin-film lithium niobate," Opt. Express **29**(23), (2021).
20. G. Yang, A. V. Sergienko, and A. Ndao, "Tunable polarization mode conversion using thin-film lithium niobate ridge waveguide," Opt Express **29**(12), 18565-18571 (2021).
21. R. Gan, L. Qi, Z. Ruan, J. Liu, C. Guo, K. Chen, and L. Liu, "Fabrication tolerant and broadband polarization splitter-rotator based on adiabatic mode evolution on thin-film lithium niobate," Opt. Lett. **47**(19), 5200-5203 (2022).
22. Z. Lin, Y. Lin, H. Li, M. Xu, M. He, W. Ke, H. Tan, Y. Han, Z. Li, D. Wang, X. S. Yao, S. Fu, S. Yu, and X. Cai, "High-performance polarization management devices based on thin-film lithium niobate," Light Sci Appl **11**(1), 93 (2022).
23. P. Zhang, H. Huang, Y. Jiang, X. Han, H. Xiao, A. Frigg, T. G. Nguyen, A. Boes, G. Ren, Y. Su, Y. Tian, and A. Mitchell, "High-speed electro-optic modulator based on silicon nitride loaded lithium niobate on an insulator platform," Opt. Lett. **46**(23), 5986-5989 (2021).
24. D. Guo and T. Chu, "Compact broadband silicon 3 dB coupler based on shortcuts to adiabaticity," Opt. Lett. **43**(19), 4795-4798 (2018).
25. H.-C. Chung, C.-H. Chen, Y. Hung, Jr., and S.-Y. Tseng, "Compact polarization-independent quasi-adiabatic 2×2 3 dB coupler on silicon," Opt. Express **30**(2), (2022).

26. H. Nikbakht, M. T. Khoshmehr, B. van Someren, D. Teichrib, M. Hammer, J. Forstner, and B. I. Akca, "Asymmetric, non-uniform 3-dB directional coupler with 300-nm bandwidth and a small footprint," Opt. Lett. **48**(2), 207-210 (2023).
27. "COMSOL Multiphysics® v. 6.1. www.comsol.com. COMSOL AB, Stockholm, Sweden.", retrieved.
28. J. Čtyroký, "3-D Bidirectional Propagation Algorithm Based on Fourier Series," J. Lightwave Technol. **30**(23), 3699-3708 (2012).
29. Z. J. Yu, X. Xi, J. W. Ma, H. K. Tsang, C. L. Zou, and X. K. Sun, "Photonic integrated circuits with bound states in the continuum," Optica **6**(10), 1342-1348 (2019).
30. J. Čtyroký, J. Petráček, V. Kuzmiak, and I. Richter, "Bound modes in the continuum in integrated photonic LiNbO3 waveguides: are they always beneficial?," Opt. Express **31**(1), 44-55 (2023).